



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Executive Summary

The research aims to identify factors contributing to successful rocket landings. Key methodologies included:

- Data collection via SpaceX REST API and web scraping.
- Data wrangling to define success/failure outcomes.
- Exploratory analysis using visualizations to assess payload, launch site, flight number, and yearly trends.
- Statistical analysis with SQL to calculate payload metrics and success rates.
- Investigation of launch site success rates and proximity to geographic markers.
- Visualization of successful launch sites and payload ranges.
- Development of predictive models (logistic regression, SVM, decision tree, KNN) to forecast landing outcomes.

Executive Summary

Results:

Exploratory Data Analysis

- Launch success rates have increased over time.
- The KSC LC-39A launch site has the highest success rate.
- Orbits ES-L1, GEO, HEO, and SSO show a 100% success rate.

Visualization and Analytics

- Most launch sites are located near the equator and close to the coast.

Predictive Analytics

- All predictive models performed comparably on the test set, with no discernable difference in accuracy.

Introduction

Background

SpaceX, a pioneering space industry leader, aims to make space travel more affordable and accessible. Its achievements include delivering spacecraft to the International Space Station, deploying a satellite constellation for global internet access, and conducting manned space missions. SpaceX's innovative reuse of the Falcon 9 rocket's first stage reduces launch costs to approximately \$62 million, compared to \$165 million or more for competitors who cannot reuse rocket stages. Predicting whether the first stage will successfully land is key to estimating launch costs. This can be achieved using public data and machine learning models to forecast the likelihood of reuse for SpaceX or other providers.

Questions

- What is the nature and extent of the available data about SpaceX Falcon 9 first stage landings?
- Which machine learning model would work best (have the highest accuracy) to predict the outcome of a Falcon 9 first stage landing from a future launch?
- Will a future Falcon 9 first stage landing be successful?

Section 1

Methodology

Methodology

Executive Summary

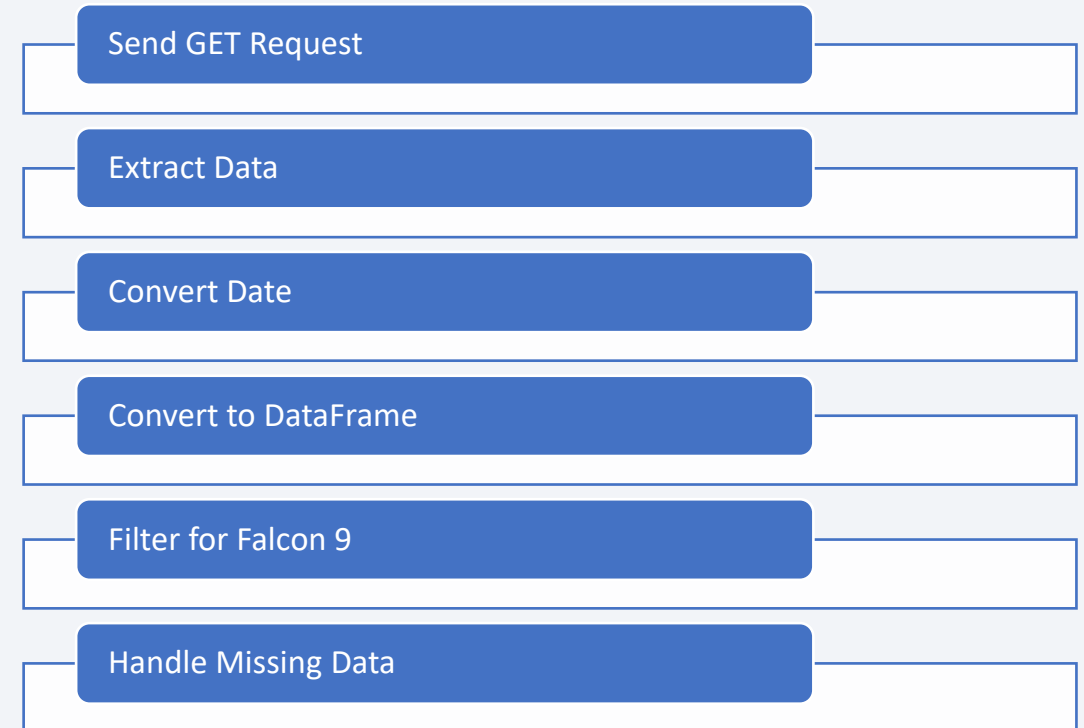
- Data collection methodology:
 - Data was collected via public APIs and web scraping. To ensure project integrity and consistent results, some data was provided with the course.
- Perform data wrangling
 - Data was filtered to keep only that which was relevant to the analysis. Missing data was either eliminated or extrapolated using various methods.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection

- Data was collected using the following methods:
 - SpaceX API
 - Data was collected by sending a GET request to SpaceX API. The data was decoded as json read into a DataFrame
 - The data was cleaned, missing values were identified and filled where necessary
 - Web-Scraping
 - Falcon 9 launch records were also collected by web-scraping Wikipedia
 - A HTTP GET request was sent to Falcon 9 launch HTML page
 - The data was parsed and read into a DataFrame using BeautifulSoup

Data Collection – SpaceX API

- SpaceX data was available at the public API endpoint:
<https://api.spacexdata.com/>
- Data was extracted from the API response read into a Pandas DataFrame for additional analysis



Source: [Data Collection](#)

Data Collection – Web Scraping

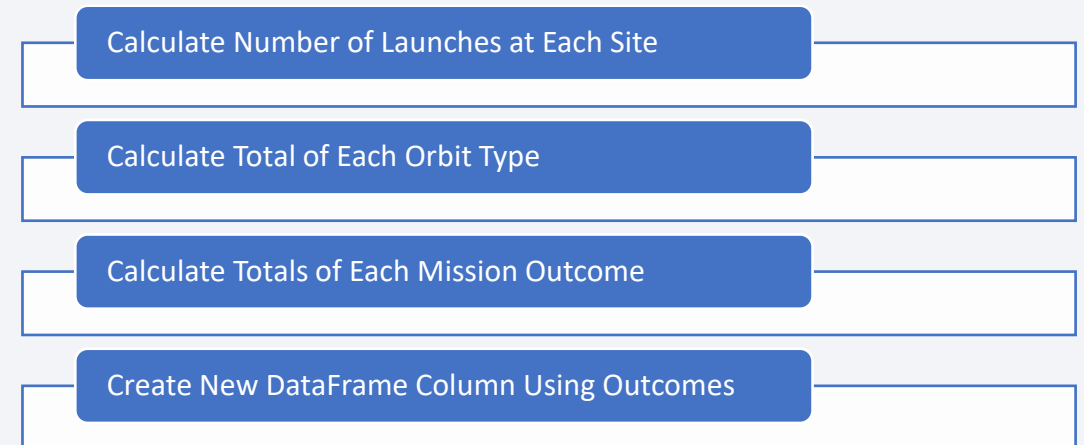
- SpaceX launch data was scraped from HTML tables on a permanently-linked copy of the SpaceX Wikipedia webpage (<https://en.wikipedia.org/wiki/SpaceX>)
- Data was extracted from the API response read into a Pandas DataFrame for additional analysis



Source: [Web Scraping](#)

Data Wrangling

- The launch sites, orbit types and mission outcomes were processed and reformatted
- The mission outcome types were converted to a binary classification (one hot encoding) where 1 represented the Falcon 9 first stage landing being a success and 0 represented a failure
- The new mission outcome classification column was added to the DataFrame



EDA with Data Visualization

The following visualizations were used:

- Launch Site
 - Scatter plot of Mission Outcome by Launch Site and Flight Number
 - Scatter plot of Mission Outcome by Launch Site and Payload
- Orbit Type
 - Bar chart of Mission Outcome vs Orbit Type
 - Scatter plot of Mission Outcome by Orbit Type and Flight Number
 - Scatter plot of Mission Outcome by Orbit Type and Payload
- Time
 - Line plot of Mission Outcome by Year

EDA with SQL

SQL queries were used to obtain the following data:

- Launch Sites
- Payload Masses
- Dates
- Booster Types
- Mission Outcomes

Build an Interactive Map with Folium

A Folium map was created with the following information:

- Markers for Johnson Space Center and all launch sites
- Circles for the launch sites
- Markers for Success/Failure
- Proximity Lines for nearby landmarks

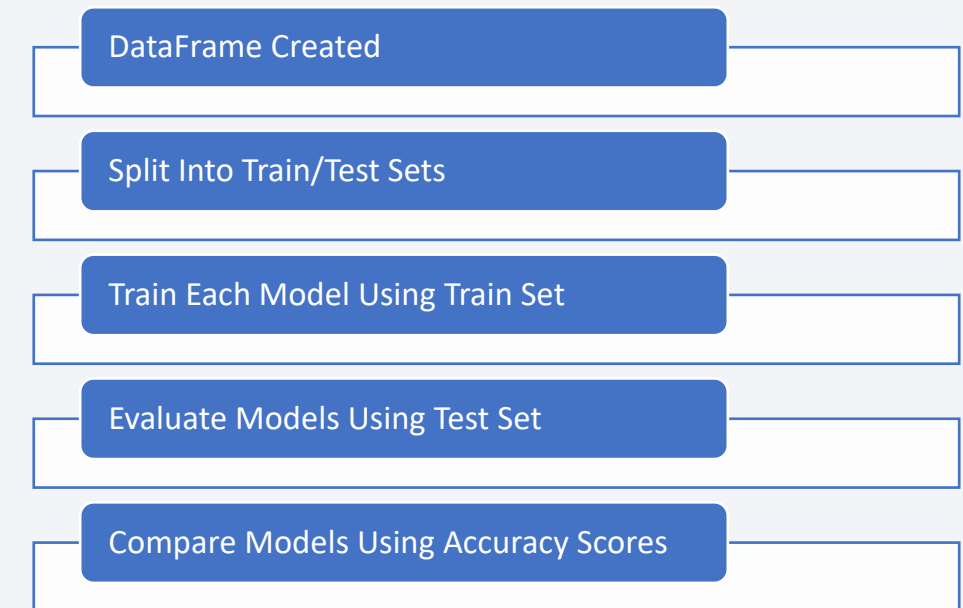
Build a Dashboard with Plotly Dash

A Plotly Dashboard was created to show the following information:

- Pie Chart
 - Dropdown selector allows selection of all launch sites or a single site
 - For All Sites, the pie chart shows the share of successful landing for each site
 - For a single site, the pie chart shows the success/fail rate for that site
- Scatter Plot
 - Displays distribution of first stage landings by payload, mission outcome, and booster version
 - Slider component allows selection of different payload ranges

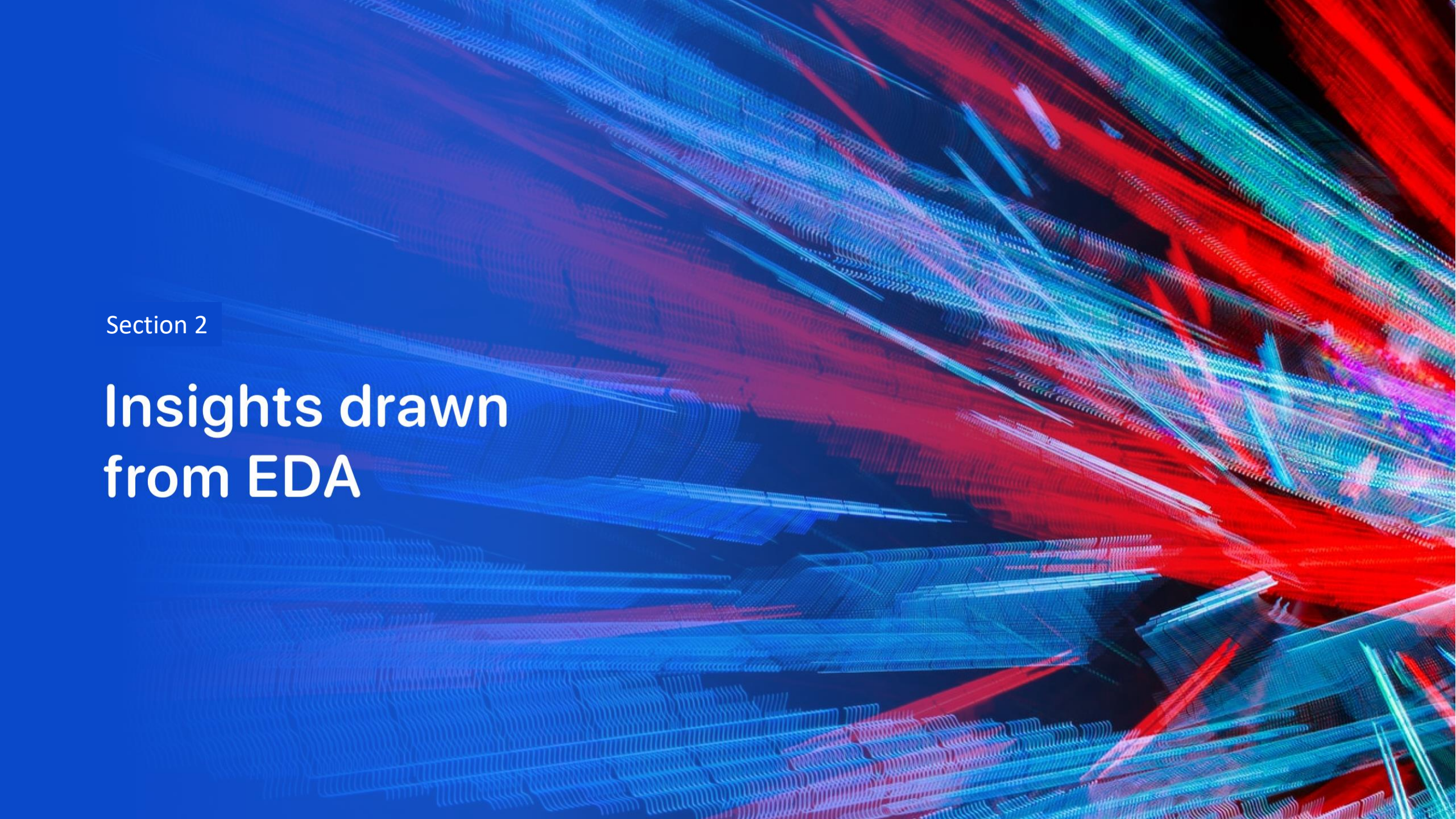
Predictive Analysis (Classification)

- Data set split into training and test sets
- The following models were trained on the training data:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)
- Hyper-parameters were evaluated using GridSearchCV
- Using the best parameters, each model was scored for accuracy using the test data sets



Results

- Exploratory data analysis results
 - Success rates have improved over time
 - CCAFS SLC-40 was the launch site that had the highest Falcon 9 first stage landing success rate (42.9%)
 - Orbits ES-L1, GEO, HEO and SSO have a 100% success rate
- Interactive analytics demo in screenshots
 - Launch sites are close to the coast but far enough from other landmarks to present a hazard
- Predictive analysis results
 - All models performed with similar accuracy

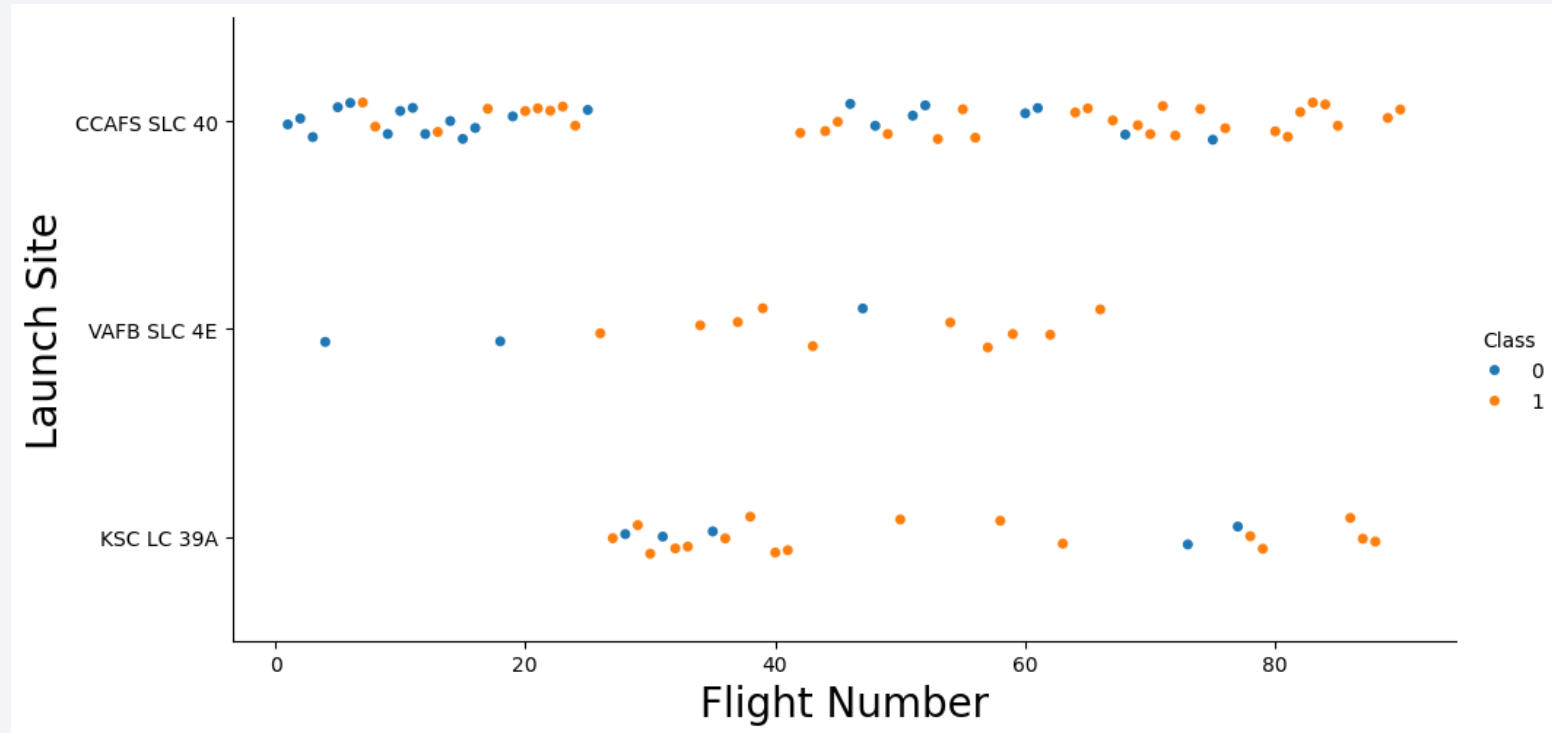
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

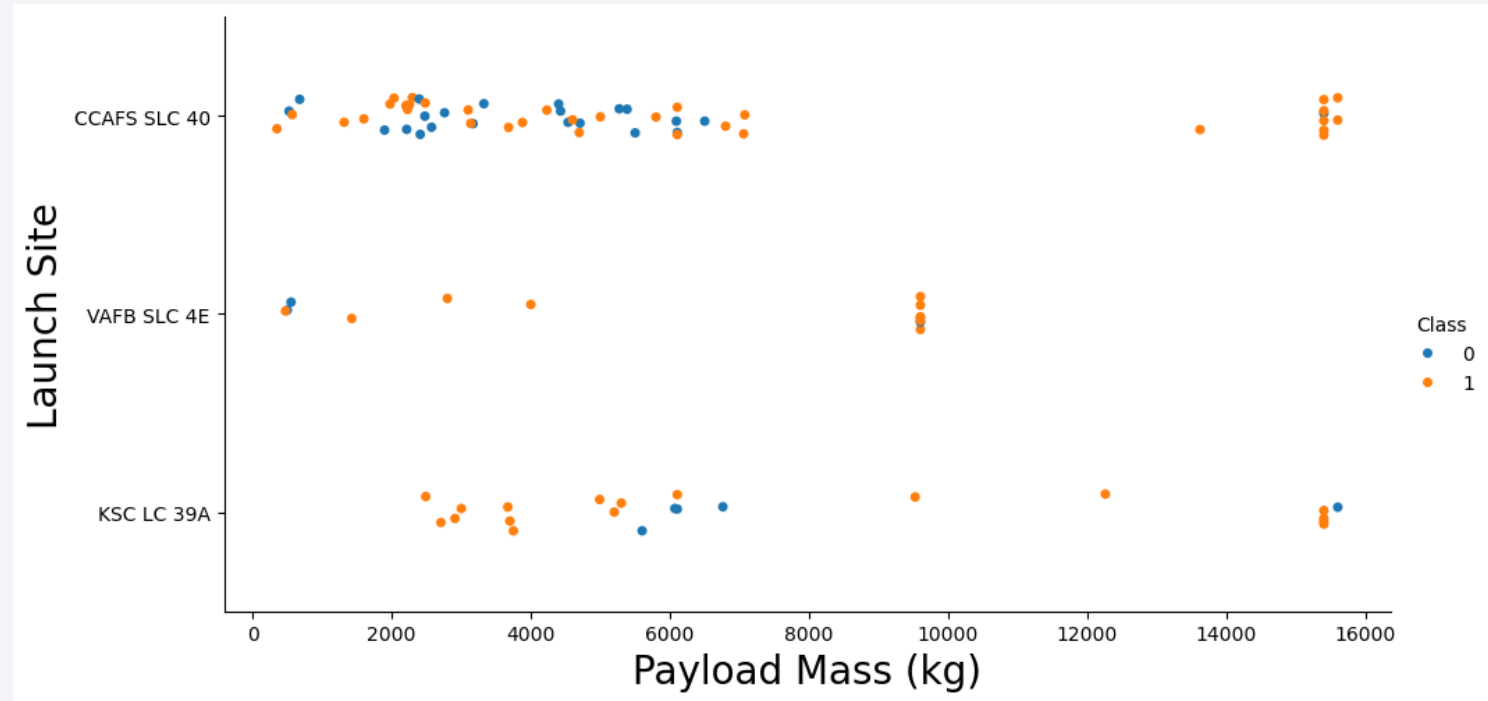
Flight Number vs. Launch Site

- This plot shows success (1) and failure (0) by launch site and flight number
- Success rate appears to increase with flight number



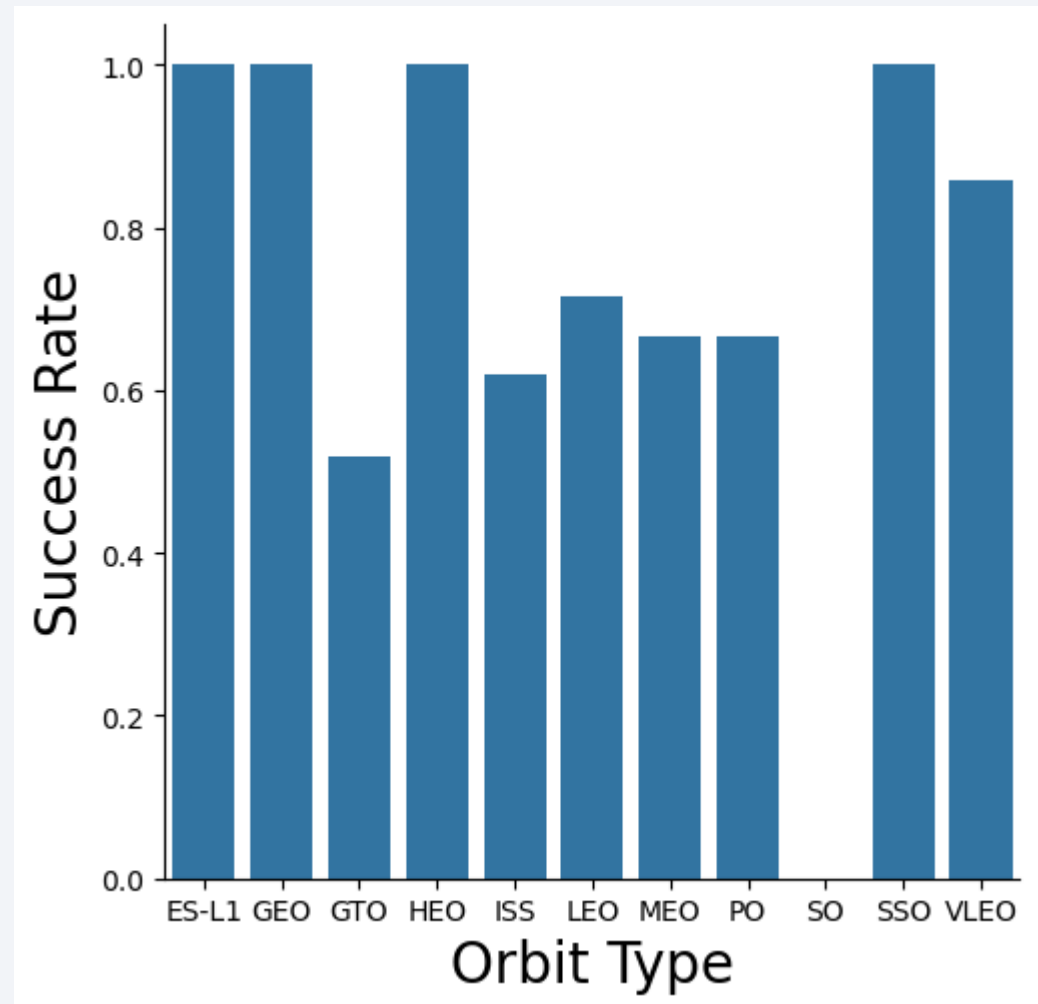
Payload vs. Launch Site

- This plot shows success and failure by launch site and payload mass
- CCAFS SLC 40 and VAFB SLC 4E show successes grouped around 16K kg and 10K kg respectively
- KSC LC 39A failures appear to mainly occur between 5500 – 7000 kg.



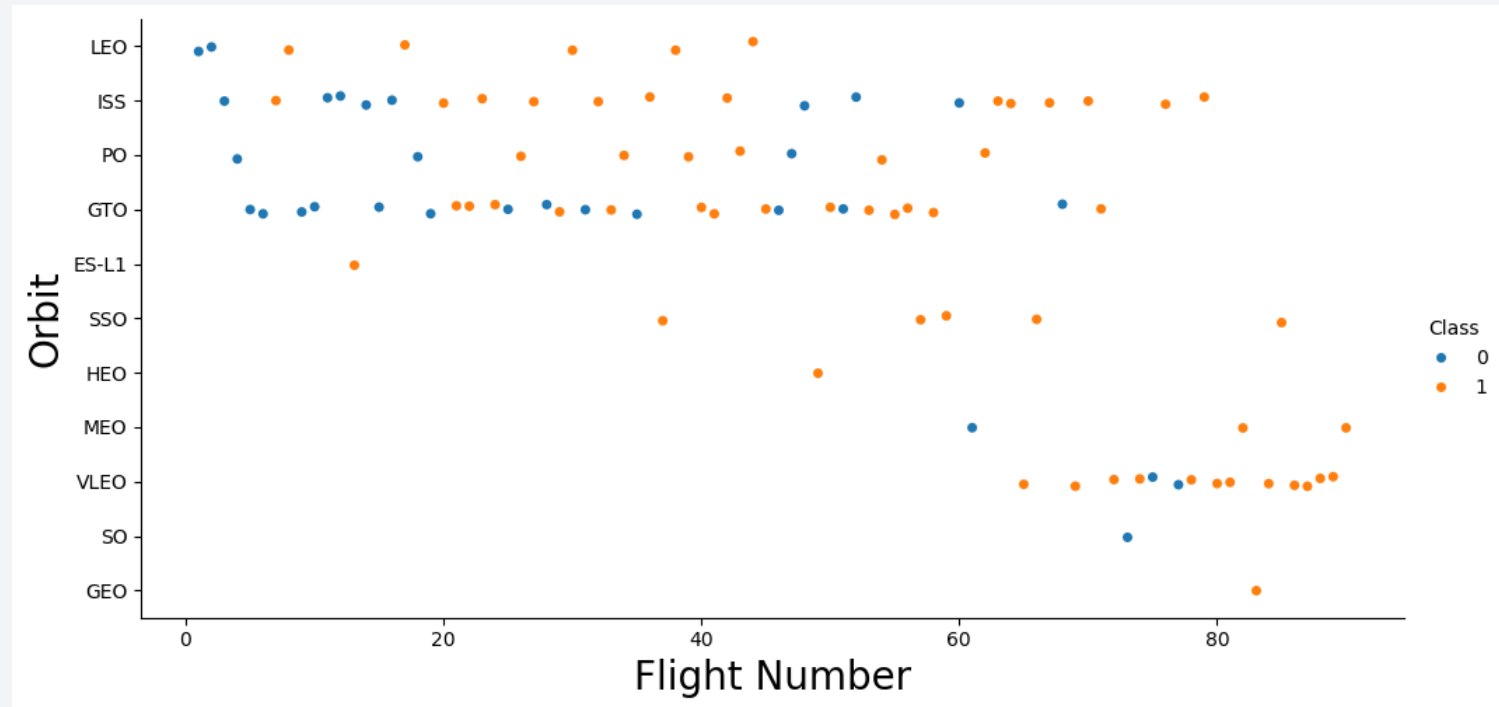
Success Rate vs. Orbit Type

- This chart shows success rate versus orbit type
- ES-L1, GEO, HEO, and SSO orbits have had no failures
- The lowest success rate is for the GTO orbit type



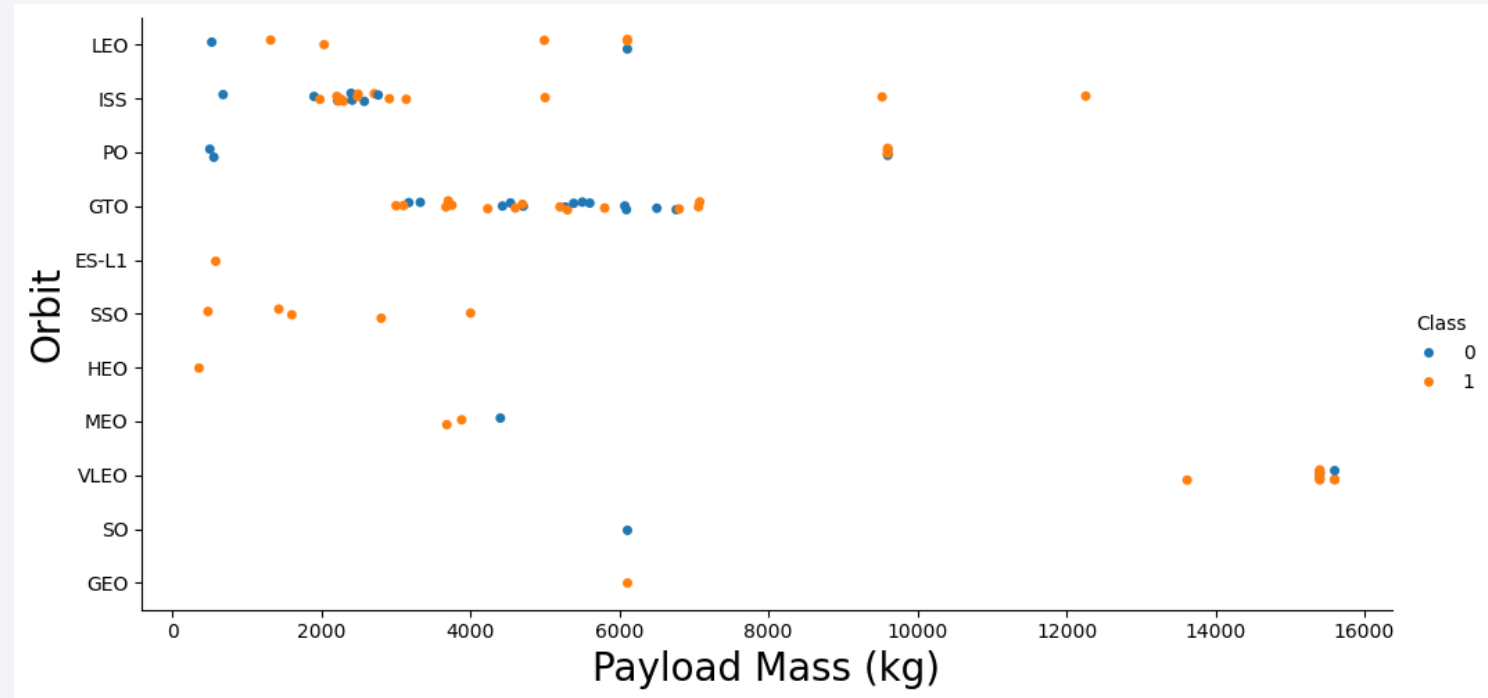
Flight Number vs. Orbit Type

- This plot shows success/failure by Flight number vs. Orbit type
- Success rate appears to increase with flight number



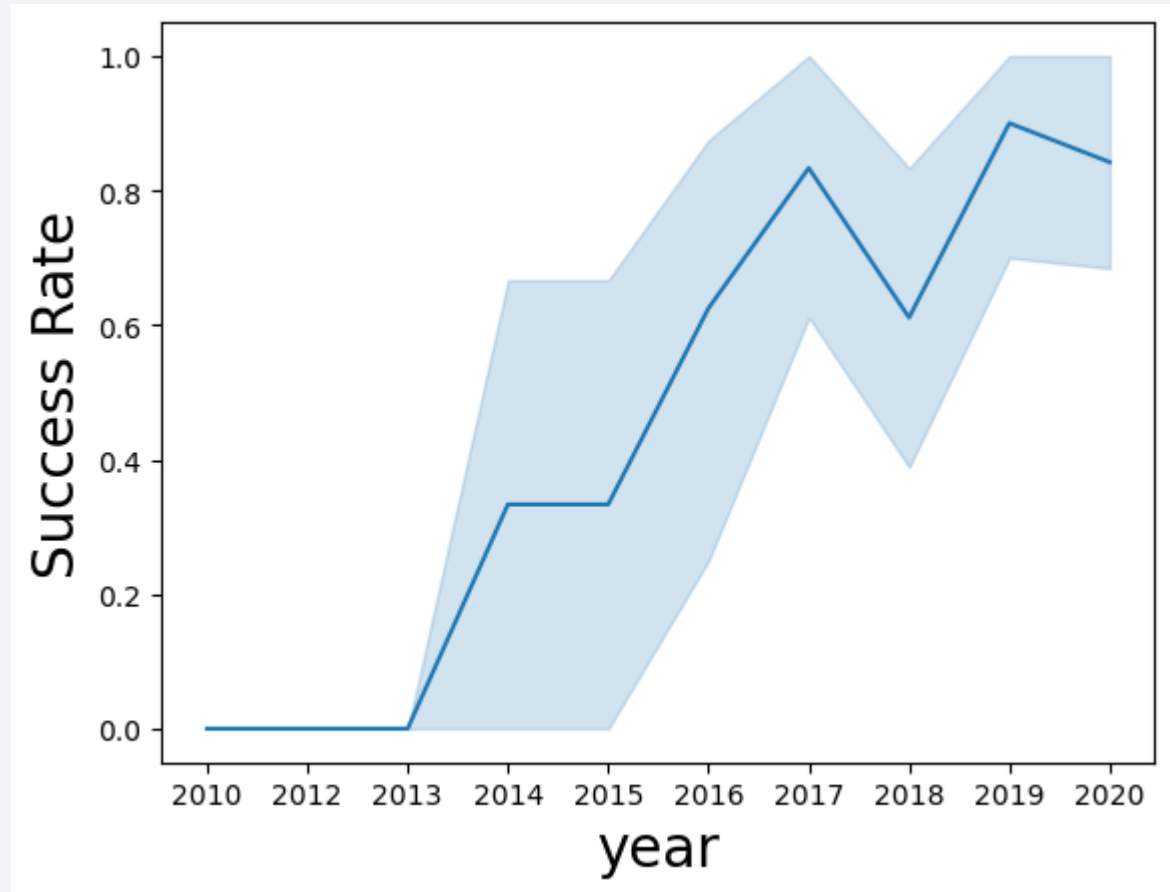
Payload vs. Orbit Type

- This plot shows success/failure by orbit vs. payload mass
- Certain orbits show higher success rates
- There does not appear to be a consistent correlation between success and payload



Launch Success Yearly Trend

- This is a plot of success rate vs year
- Success rates improve over time



All Launch Site Names

- Find the names of the unique launch sites

Launch Sites

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

Total Payload Mass (kg)

45596

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

Average Payload Mass (kg)

2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

Date of first success

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

Mission Outcome	Count
Failure	1
Success	100

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- List the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

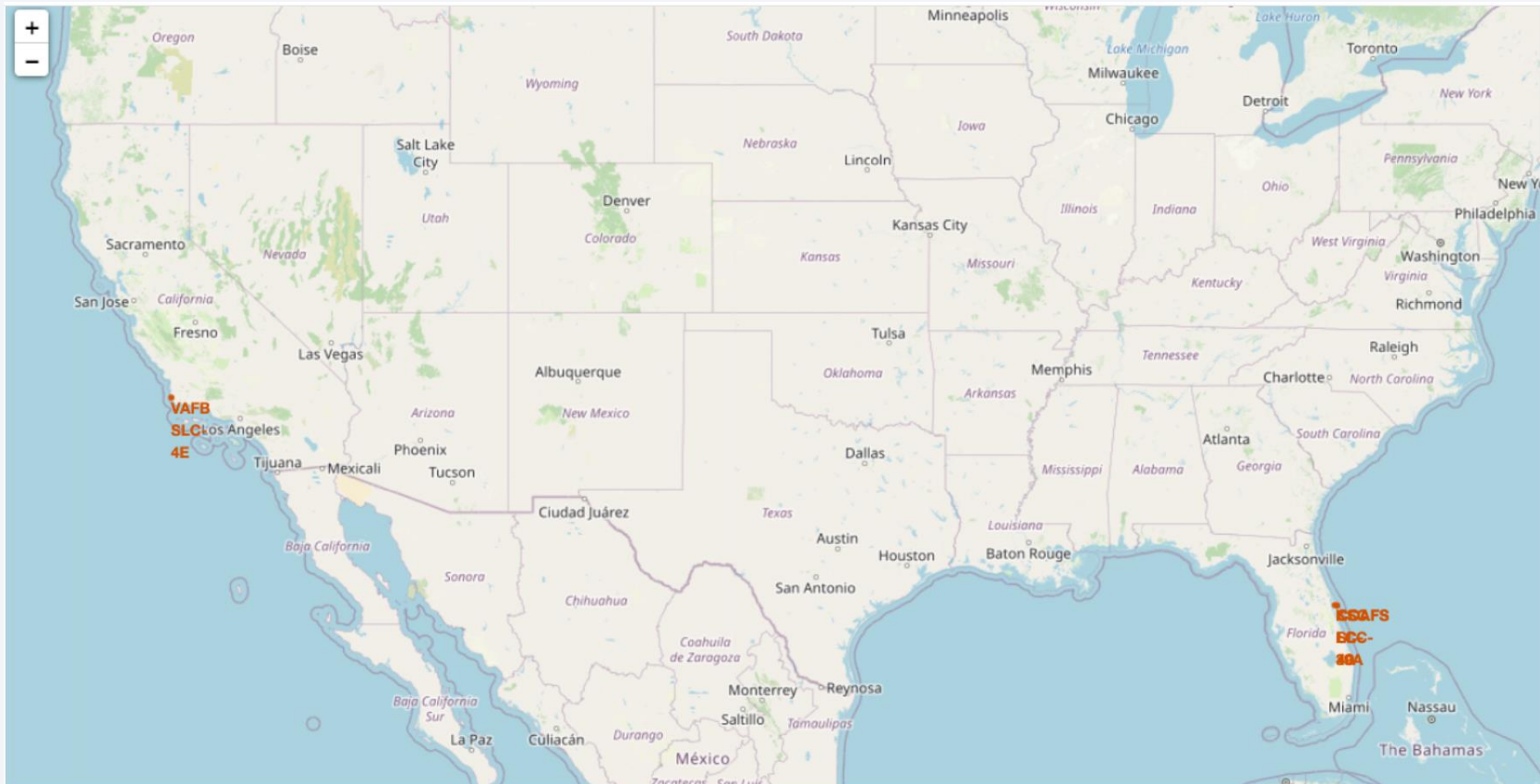
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

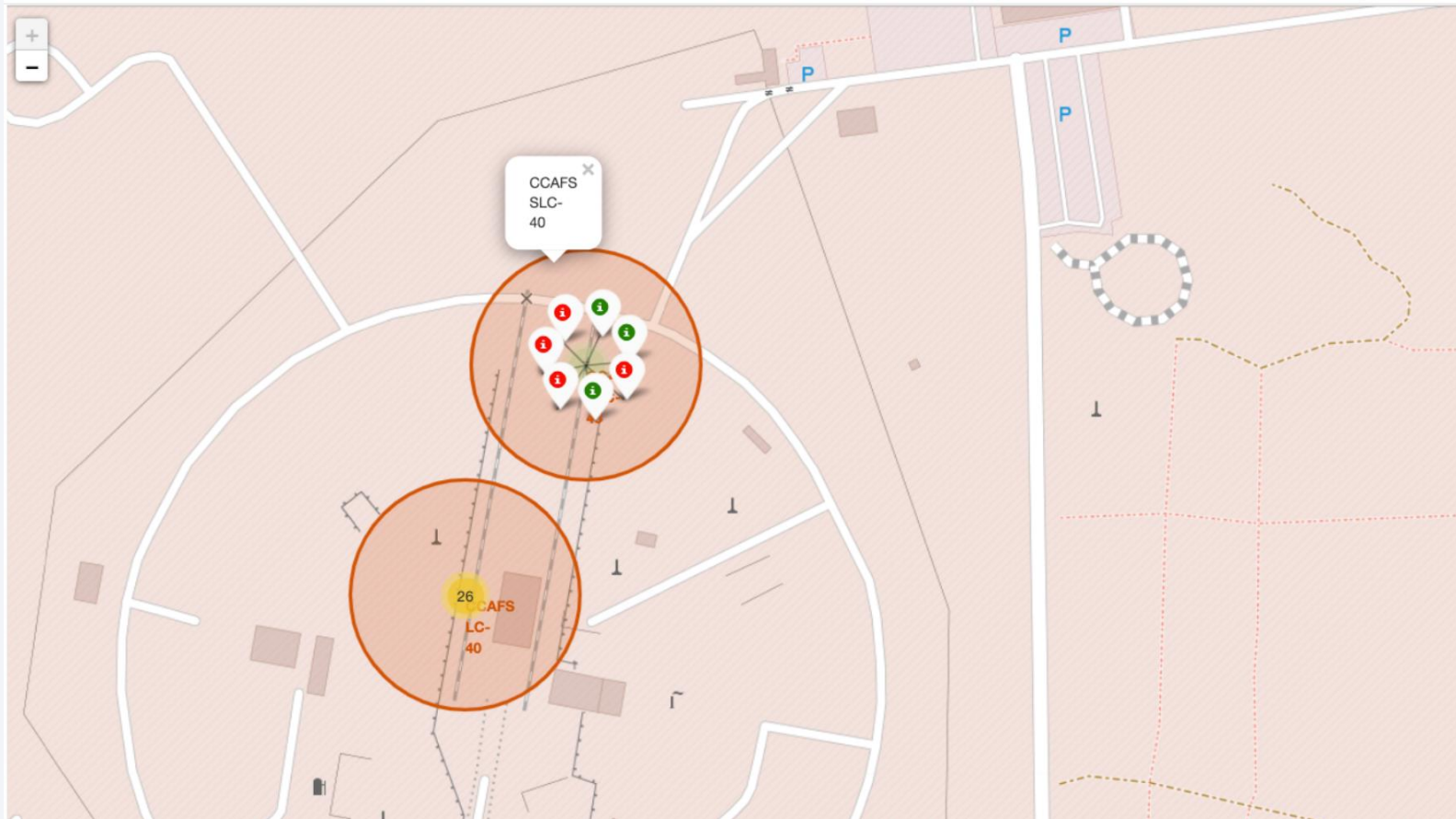
Map of Launch Sites

- Location of Launch Sites in the United States



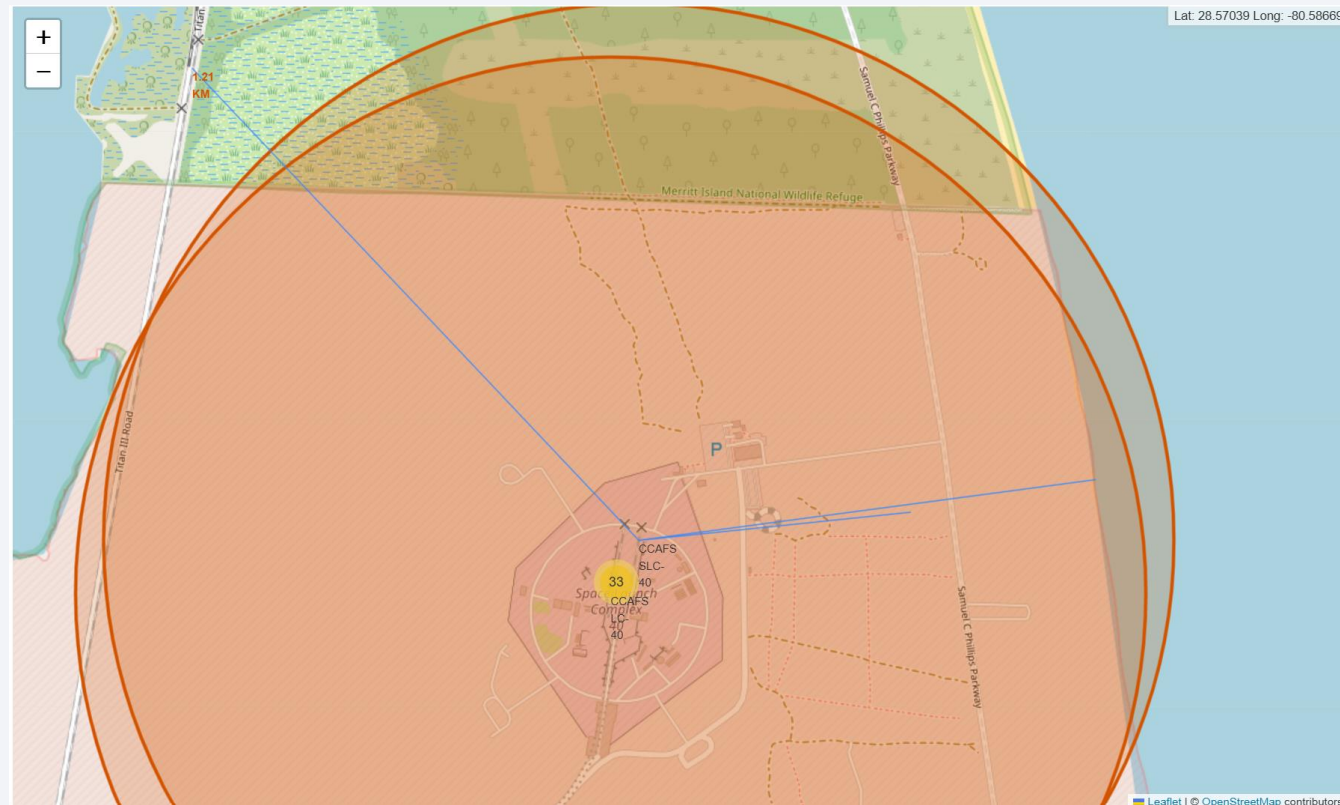
Launch Outcomes

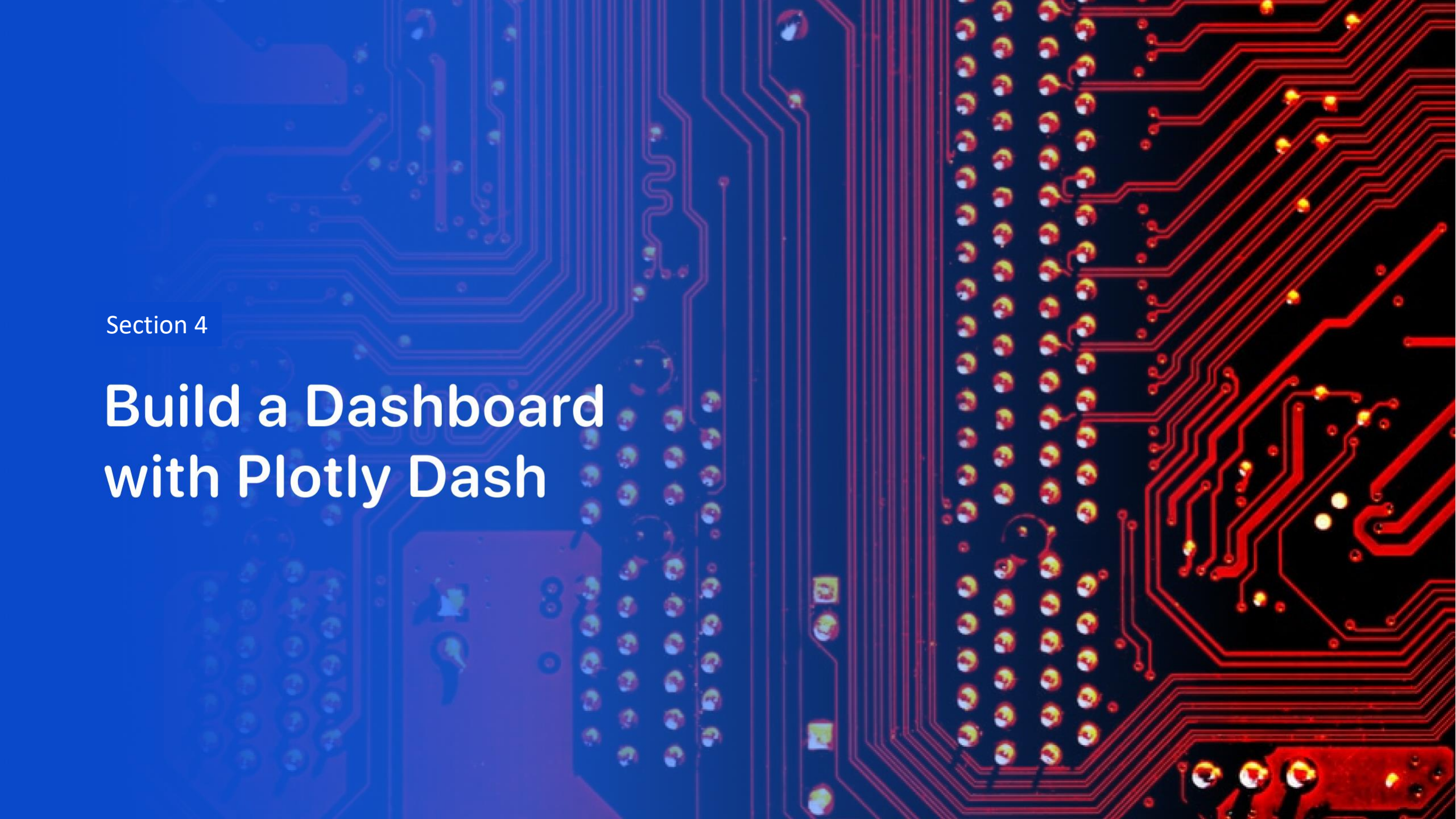
- Launch outcomes for CCAFS SLC-40



Launch Site Landmark Proximity

- Launch site CCAFS SLC-40 proximity to coast and railroad



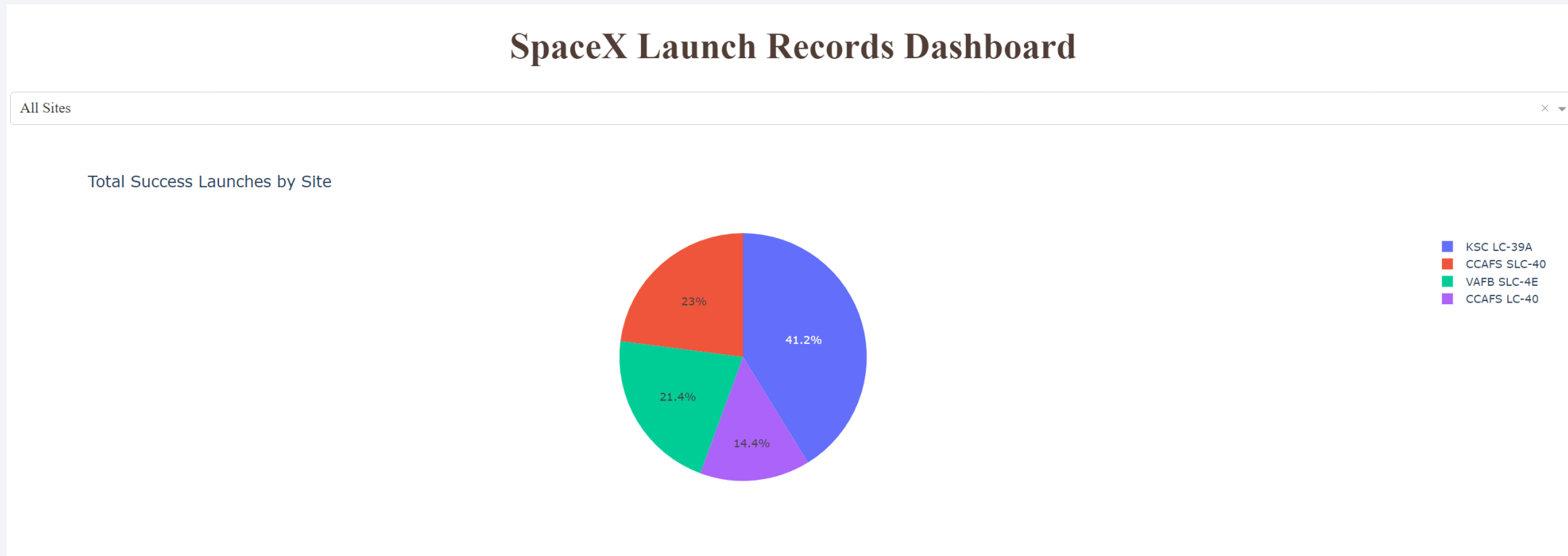


Section 4

Build a Dashboard with Plotly Dash

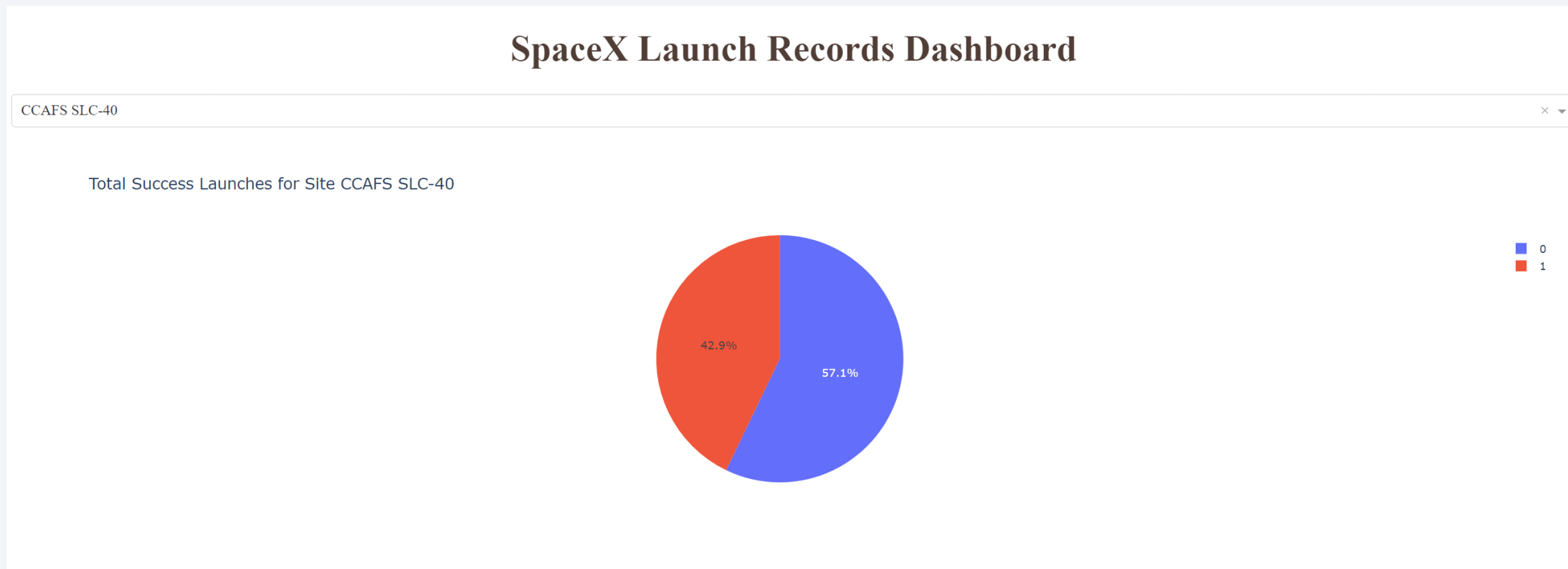
Success Rate by Launch Site

- KSC LC-39A launched have the highest success rate for landings



Launch Site With Highest Success Rate

- CCAFS SLC-40 has had the highest landing success rate



Success Correlation with Payload Mass and Booster Version

- Success data for payloads between 1000 kg and 8000 kg



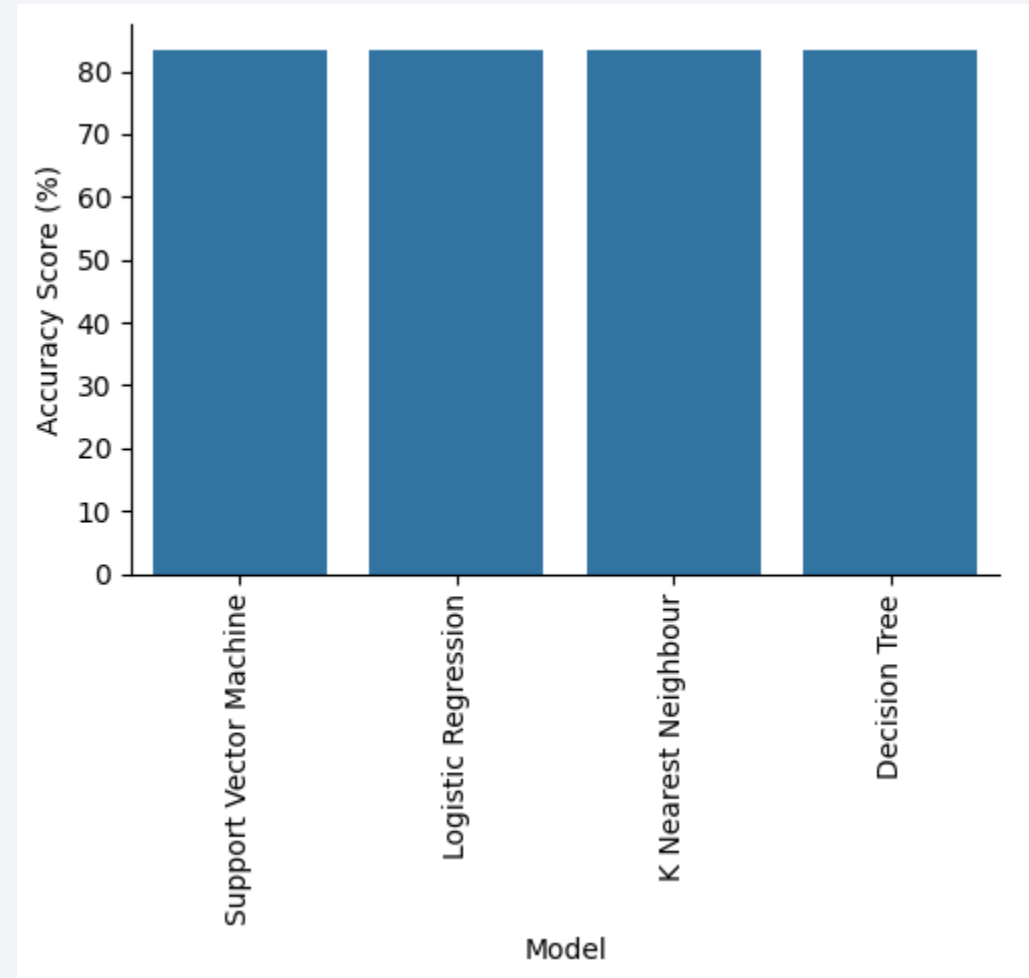


Section 5

Predictive Analysis (Classification)

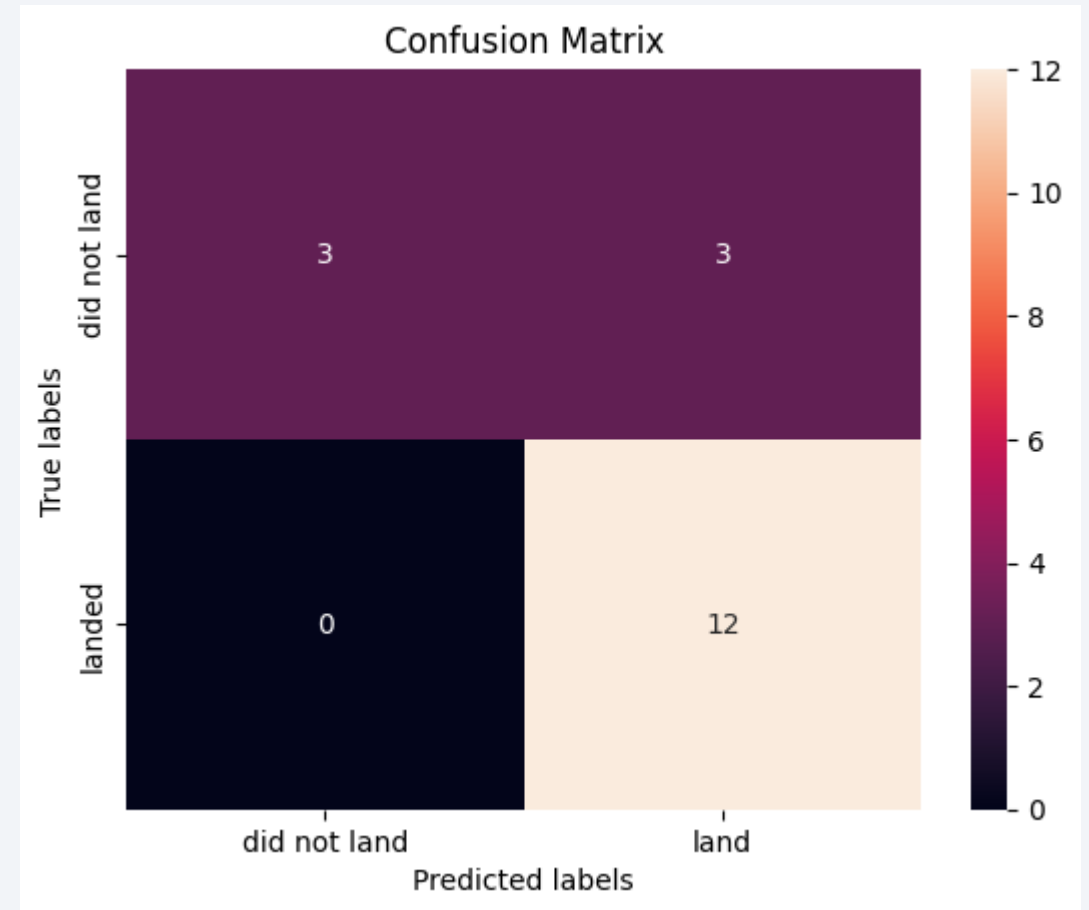
Classification Accuracy

- Accuracy scores for each model
- All models had similar accuracy



Confusion Matrix

- This shows the confusion matrix for the Logistic Regression Model
- Results
 - 12 True Positives (lower right corner)
 - 3 True Negatives (upper left corner)
 - 0 False Negatives (lower left corner)
 - 3 False Positives (upper right corner)



Conclusions

- Landing success rate is increasing over time
- Some orbit types are correlated with higher landing success
- Higher payload masses are correlated with higher landing success
- All models show similar accuracy in predicting future success.
- Additional data should be collected to improve the models and increase prediction accuracy

Appendix

- Github project: [IBM Data Science Capstone Project](#)

Thank you!

